

Jerry Peng

jp2011pa@gmail.com | 484-999-1662 | [linkedin.com/in/jerry-peng-706711231](https://www.linkedin.com/in/jerry-peng-706711231) | peng-jerry.github.io

Education

Carnegie Mellon University | College of Engineering

Expected Graduation: December 2026

M.S. in Mechanical Engineering | Concentration Area: Robotic & Control Systems | GPA 4.0/4.0

Relevant Coursework: Advanced Control Systems Integration, Optimal Control & Reinforcement Learning, Modern Control Theory, Robot Dynamics, ML/AI for Engineers, Advanced Mechatronic Design, Electro-Mechanical Systems Design

University of Pittsburgh | Swanson School of Engineering

May 2025

B.S. in Mechanical Engineering; Honors Degree | GPA 3.9/4.0

Work Experience

Boat Project | Robomechanics Lab

Pittsburgh, PA

Graduate Research Assistant

September 2025 - Present

- Developed MOSAIC-SV, a UKF & MPPI-based dynamics estimator and controller for unmanned surface vessels (USVs), designed as a data-efficient alternative to traditional USV control architectures, presented at IROS 2026 under the Perception and Navigation for Marine Robotics Workshop ([OpenReview](#))
- Improved transit time by 10.1% over the pre-derived Fossen base model under static model mismatch, while maintaining robust performance under external current disturbances

Magnetic Wall Climber | Robomechanics Lab

Pittsburgh, PA

Graduate Research Assistant

April 2026 – Present

- Designed a 4-DOF IK solver combining analytic chain solving with BFGS refinement to determine optimal joint angles for floor, wall, ceiling, and thin-edge transitions
- Engineered constraint and smoothness-weighted cost functions to eliminate joint-angle discontinuities and wheel-velocity spikes near kinematic singularities during surface transitions
- Built the software-to-hardware pipeline, driving motor (Westwood Robotics BEAR) via custom homing, calibration, and closed-loop position-control routines for real-time trajectory playback

ESTAT Actuation

Pittsburgh, PA

Mechanical Engineering Co-op & Rotary Lead

January 2024 - August 2024

- Led mechanical design on over 20 custom rotary clutch designs culminating in a catalog featured at the 2024 Automate Show
- Refined the rotary brake development process cutting design time by over 50% and saving 100+ hours in product development
- Coordinated with Suzuki on brake design for the Moqba 2 featured at the 2025 Japan Mobility Show by providing design documentation and performing reliability testing, helping secure a purchase order of over \$100,000

Project Experience

Non-linear Regulation & Robustness | Carnegie Mellon University

January 2026 - April 2026

- Derived LQR, partial feedback linearization, and sliding mode controllers in Simulink and implemented them on an underactuated rotary inverted pendulum (Quanser Qube Servo 2)
- Validated controller robustness on hardware by injecting 20% parameter uncertainty and external disturbances, demonstrating SMC's superior disturbance rejection over LQR and PFL

Launch-a-Birdie | Carnegie Mellon University

August 2025 - December 2025

- Designed and built a low-cost robotic badminton launcher to enable high-volume and consistent shot practice
- Engineered a full electrical system for the 7-motor mechatronic setup, including power distribution, ESC & motor driver integration, and Arduino control, allowing for real-time coordination between subsystems, achieving a <1m deviation in all shot types

Skills & Certifications

Control: Simulink, PID, LQR, MPPI, Adaptive Control, Sliding Mode Control, Kalman Filtering, USART & I2C Communication

Programming: Python, MATLAB, C, C++ (Arduino IDE), ROS, Linux, Git

Embedded: Arduino, Microcontrollers (ATmega328P), Oscilloscopes & Multimeters, USART & I2C Communication, PWM

Design: CAD (Solidworks), Simulation (Solidworks, COMSOL, UE5), PCB Design (Altium Designer, KiCAD), Circuit Design

Technical Expertise: FMEA, DFM, Stress Analysis, Parts Sourcing, Engineering Documentation, GD&T, Rapid Prototyping

Prototyping: Laser Cutter (ULS PLS6.75), Waterjet (Wazer), CNC Mill, Manual Lathe & Mill, Band Saw, 3D-Printing, Soldering

Certifications: FE Exam (Mechanical) - Passed July 2026